Random counterexamples to three Grundy product conjectures

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Abstract

We disprove the hypergraph expanding-sequence product conjecture and, by the standard incidence-graph translations, the Grundy total-domination direct-product and Grundy domination strong-product conjectures. A reflexive random tournament matrix of order 1937 has, with positive probability, no triangular witness of length 45. Its product with its transpose nevertheless contains a 1937×1937 identity submatrix. Translating this single probabilistic object gives a connected co-bipartite graph counterexample and an incidence bipartite graph counterexample. No explicit order-1937 matrix is claimed, and we make no priority claim.

1 Introduction

The Grundy domination number is the maximum length of a legal sequence of distinct vertices, where each term has a vertex in its closed neighborhood not dominated by earlier terms. The strong-product conjecture of Brešar et al. [4] predicts $\gamma_{\mathrm{gr}}(G \boxtimes H) = \gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$; the analogous total-domination conjecture predicts $\gamma^t_{\mathrm{gr}}(G \times H) = \gamma^t_{\mathrm{gr}}(G)\gamma^t_{\mathrm{gr}}(H)$. The hypergraph formulation and equivalence of these three conjectures were given in [2] (published as [3]). We provide a counterexample to all three statements. The construction repurposes the classical random fooling-set/tournament and $A \otimes A^T$ mechanisms [10, 7] for the expanding-sequence parameter. Neither the random model nor the identity-block mechanism is new, and we make no priority claim.

2 Expanding sequences and matrices

For a 0–1 matrix A, write $A \otimes B$ for the Kronecker product.

Definition 2.1. *For ordered injective row indices $r_1, \dots, r_k$ and column indices $c_1, \dots, c_k$, call the pair a triangular witness if*

$$A[r_i, c_i] = 1, \qquad A[r_i, c_j] = 0 \quad (i < j).$$

Let $\tau(A)$ be the largest possible k.

Viewing the rows as hyperedges on the column set, the condition says that each new hyperedge contains a point not in the earlier hyperedges, with the chosen private points ordered by the sequence. Thus $\tau(A)$ is exactly the maximum expanding-row (or expanding-hyperedge) sequence length. In graph-product terminology, the same matrix pattern is also called a semi-induced matching; this should not be silently identified with every definition of the standard semi-ladder index [5, 6].

Lemma 2.2. $\tau(A^T) = \tau(A)$.

Proof. Transpose a triangular witness and reverse both orders. The diagonal-one conditions remain diagonal-one. A zero $A[r_i, c_j]$ with $i < j$ becomes the zero $A^T[c_j, r_i]$, which is above the diagonal after reversing the two orders. The operation is an involution. □

3 The random theorem

Let M be the reflexive random tournament matrix of order n: its diagonal is one, and for every unordered pair $\{u, v\}$ exactly one of $M[u, v]$ and $M[v, u]$ is one, independently across pairs.

Lemma 3.1. *For fixed injective ordered row and column k-tuples,*

$$\Pr[\textit{they form a triangular witness}] \le 2^{-\binom{k}{2}}.$$

Proof. There are $\binom{k}{2}$ upper-zero constraints. If one is a diagonal cell, the event is impossible because diagonal entries are one. Two distinct constraints cannot use the same directed cell, since both tuples are injective. If two constraints use the same unordered tournament edge in opposite directions, they require both orientations to be zero, again making the event impossible. Therefore, whenever the event is nonempty, all $\binom{k}{2}$ zero constraints use distinct independent tournament bits. The diagonal-one constraints can only reduce the event. Hence its probability is at most $2^{-\binom{k}{2}}$. □

Theorem 3.2. *For $n = 1937$ there exists a reflexive tournament matrix M with $\tau(M) \le 44$ and*

$$\tau(M \otimes M^T) \ge 1937.$$

Proof. Union bound over the $(n)_k^2$ ordered injective row and column tuples gives

$$\Pr[\tau(M) \ge k] \le (n)_k^2 2^{-\binom{k}{2}} < n^{2k} 2^{-\binom{k}{2}}.$$

For $n = 1937$, $k = 45$, $n < 2^{11}$ and $\binom{45}{2} = 990$, so the last quantity is less than

$$(2^{11})^{90} 2^{-990} = 2^0 = 1.$$

Thus some M has $t := \tau(M) \le 44$. For rows (i, i) and columns (j, j),

$$(M \otimes M^T)[(i,i),(j,j)] = M[i,j]M[j,i] = \delta_{ij}.$$

These rows and columns form an identity submatrix, whose diagonal witness has length $n = 1937$. This proves the second assertion. □

Theorem 3.2 is already a counterexample to multiplicativity of the hypergraph expanding-sequence parameter, since

$$\tau(M \otimes M^T) \ge 1937 > 44^2 \ge \tau(M)\tau(M^T).$$

4 Strong-product graph corollary

For a graph G, a closed-neighborhood sequence is legal if each selected vertex has a private vertex in its closed neighborhood outside the union of the earlier closed neighborhoods. The Grundy domination number $\gamma_{\mathrm{gr}}(G)$ is the maximum length of such a sequence. We use $N_{G \boxtimes H}[(g,h)] = N_G[g] \times N_H[h]$.

Define $K(M)$ as the co-bipartite graph with cliques $R = \{r_i : i \in [n]\}$ and $C = \{c_j : j \in [n]\}$, with cross-edge $r_i c_j$ iff $M[i,j] = 1$.

Lemma 4.1. $\gamma_{\mathrm{gr}}(K(M)) = \tau(M)$.

Proof. An R-only legal sequence is exactly an expanding-row sequence of M; similarly a C-only sequence is one of M^T. By Lemma 2.2 both have maximum length $\tau(M)$. If a legal sequence switches cliques, then after the first selected vertex from each clique both cliques are dominated, so the switch is its last term. Suppose the prefix lies in R and has length s, and let c_j be the switching term. Its private witness is a column c_x: a row-side witness is impossible because the first selected vertex in the R-clique already dominates all of R. Since c_x was not covered by the prefix, r_x was not selected (otherwise reflexivity would cover c_x), and r_x can be appended using c_x as private witness. Thus every mixed sequence can be replaced by an R-only legal sequence of at least the same length. The transpose argument handles a prefix in C. Conversely, an inclusion-maximal expanding sequence is dominating: an undominated column c_j cannot have its index among the selected rows, so the unselected row r_j is appendable; the row case is identical. Hence the maximum is $\tau(M)$. $\square$

The graph $K(M)$ is connected, co-bipartite, and has 3874 vertices. In $K(M) \boxtimes K(M)$ put $z_i = (r_i, c_i)$ and $w_j = (c_j, r_j)$. Since $N[(u,v)] = N[u] \times N[v]$,

$$w_j \in N[z_i] \quad \Longleftrightarrow \quad M[i,j]M[j,i] = \delta_{ij}.$$

Thus $z_1, \ldots, z_n$ is legal, with private witnesses $w_1, \ldots, w_n$. Any finite legal prefix extends to a dominating sequence by appending legal vertices, so

$$\gamma_{\mathrm{gr}}(K(M) \boxtimes K(M)) \geq 1937 > 44^2 \geq \gamma_{\mathrm{gr}}(K(M))^2.$$

5 Total-Grundy direct-product corollary

For a graph G without isolated vertices, an open-neighborhood sequence is legal if each term has a vertex in its open neighborhood outside the earlier open neighborhoods; the maximum length is the Grundy total-domination number $\gamma_{\mathrm{gr}}^t(G)$. The direct product has vertex set $V(G) \times V(H)$ and adjacency $(g,h) \sim (g',h')$ exactly when $gg' \in E(G)$ and $hh' \in E(H)$. The hypergraph parameter above is the corresponding expanding-edge parameter.

For a matrix A, let $B(A)$ be its incidence bipartite graph: left vertices are rows, right vertices are columns, and $i \sim j$ iff $A[i,j] = 1$.

Lemma 5.1.

$$\gamma_{\mathrm{gr}}^t(B(A)) = \tau(A) + \tau(A^T) = 2\tau(A).$$

Proof. The open neighborhoods of left-side terms lie in the right side, while those of right-side terms lie in the left side, so the two subsequences do not interfere. Each side is exactly a triangular witness, giving the first equality; Lemma 2.2 gives the second. $\square$

The four vertex types in $B(M) \times B(M)$ split into two disjoint subgraphs by whether the two coordinates lie on the same side. Directly from the incidence condition, with the displayed coordinate orderings,

$$B(M) \times B(M) \cong B(M \otimes M) \ \sqcup \ B(M \otimes M^T).$$

The first isomorphism uses (R, R) as rows and (C, C) as columns; the second uses (R, C) as rows and (C, R) as columns. Grundy total-domination is additive over disjoint unions, because open neighborhoods do not cross the union. The second factor is $M \otimes M^T$, not another copy of $M \otimes M$. The standard lexicographic witness gives $\tau(M \otimes M) \geq t^2$: take the witness pairs for M in outer order and use the witness pairs in inner order within each outer pair. The identity submatrix in Theorem 3.2 gives $\tau(M \otimes M^T) \geq n$. Therefore, using disjoint components additively,

$$\gamma^t_{\text{gr}}(B(M) \times B(M)) \geq 2t^2 + 2n > 4t^2 = \gamma^t_{\text{gr}}(B(M))^2.$$

6 Asymptotic strengthening

Fix $\varepsilon > 0$ and set $k = \lceil (4 + \varepsilon) \log_2 n \rceil$. The base-two logarithm of the union bound is

$$2k \log_2 n - \frac{k(k-1)}{2} = -(2\varepsilon + \varepsilon^2/2)(\log_2 n)^2 + O(\log n),$$

which tends to $-\infty$. Hence for all sufficiently large n some tournament matrix satisfies $\tau(M) = O(\log n)$, while $\tau(M \otimes M^T) \geq n$. The multiplicative violation is therefore $\Omega(n/\log^2 n)$. The two graph translations above inherit the same unbounded gap (up to the fixed square/product conversion).

7 Related work and scope

The strong-product conjecture originated in [4]. The total-Grundy direct-product conjecture, hypergraph formulation, and stated equivalence appear in [2, 3]. The forest-factor positive theorem is [1]. A proof attempt was later withdrawn; the arXiv record explicitly marks version 2 as withdrawn and notes an error in its proof [8].

The random fooling-set/tournament perspective and the classical tensor identity mechanism belong to the fooling-set/rank literature [10, 7]; the expanding-sequence/semi-induced- matching correspondence also appears in graph-product work [5]. Standard semi-ladder terminology is surveyed in [6] and recent complexity work [9]. Our computational package exhaustively checks only small tournaments for mutation and indexing errors; it is not evidence for the existential $n = 1937$ theorem. The literature search recorded in the file 'research/LiteratureAudit.md' does not establish priority for this counterexample.

References

[1] Kayla Bell, Keith Driscoll, Elliot Krop, and Kimberley Wolff. Grundy domination of forests and the strong product conjecture. *The Electronic Journal of Combinatorics*, 28(2):P2.12, 2021.

[2] Boštjan Brešar, Csilla Bujtás, Tanja Gologranc, Sandi Klavžar, Gašper Košmrlj, Tilen Marc, Balázs Patkós, Zsolt Tuza, and Máté Vizer. On grundy total domination number in product graphs. 2017.

[3] Boštjan Brešar, Csilla Bujtás, Tanja Gologranc, Sandi Klavžar, Gašper Košmrlj, Tilen Marc, Balázs Patkós, Zsolt Tuza, and Máté Vizer. On grundy total domination number in product graphs. *Discussiones Mathematicae Graph Theory*, 41(1):225–247, 2021.

[4] Boštjan Brešar, Csilla Bujtás, Tanja Gologranc, Sandi Klavžar, Gašper Košmrlj, Balázs Patkós, Zsolt Tuza, and Máté Vizer. Dominating sequences in grid-like and toroidal graphs. *The Electronic Journal of Combinatorics*, 23(4):P4.34, 2016.

[5] Parinya Chalermsook, Bundit Laekhanukit, and Danupon Nanongkai. Graph products revisited: Tight approximation hardness of induced matching, poset dimension and more. In *Proceedings of the Twenty-Fourth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 1557–1576, 2013.

[6] Grzegorz Fabiański, Martin Grohe, Michał Kędzierski, Juan Tan, and Szymon Toruńczyk. Progressive algorithms for domination and independence. In *36th International Symposium on Theoretical Aspects of Computer Science (STACS 2019)*, volume 126 of *LIPIcs*, pages 27:1–27:18, 2019.

[7] Mirjam Friesen, Aya Hamed, Troy Lee, and Dirk Oliver Theis. Fooling sets and rank, 2014.

[8] Rebekah Herrman and Stephen G. Z. Smith. A proof of the grundy domination strong product conjecture, 2023. Withdrawn; arXiv version 2 notes an error in the proof.

[9] Pasin Manurangsi. On the computational complexity of (extended) threshold dimension and (semi-)ladder index, 2026.

[10] Mozhgan Pourmoradnasseri and Dirk Oliver Theis. The (minimum) rank of typical fooling set matrices, 2016.